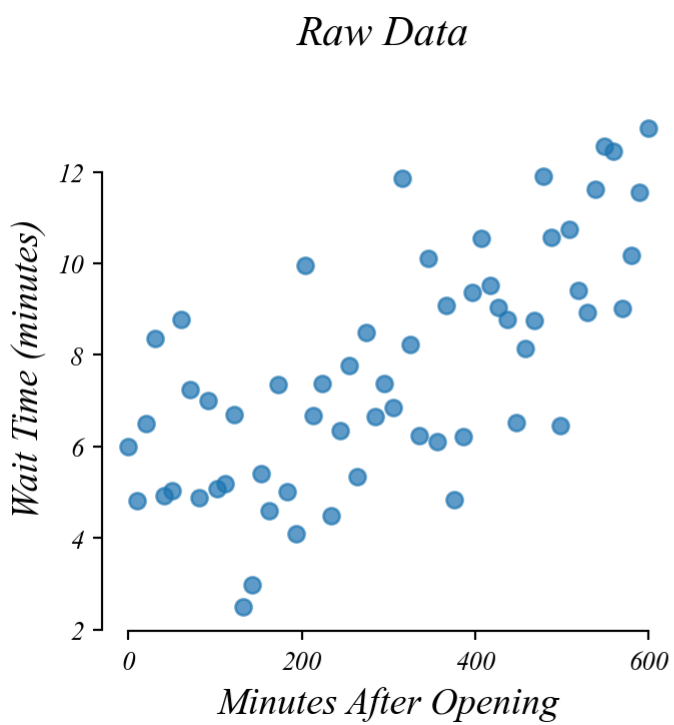
# ECON 0150 | Economic Data Analysis

*The economist's data analysis stillset.*

## *Part 4.1 | Numerical Predictors*

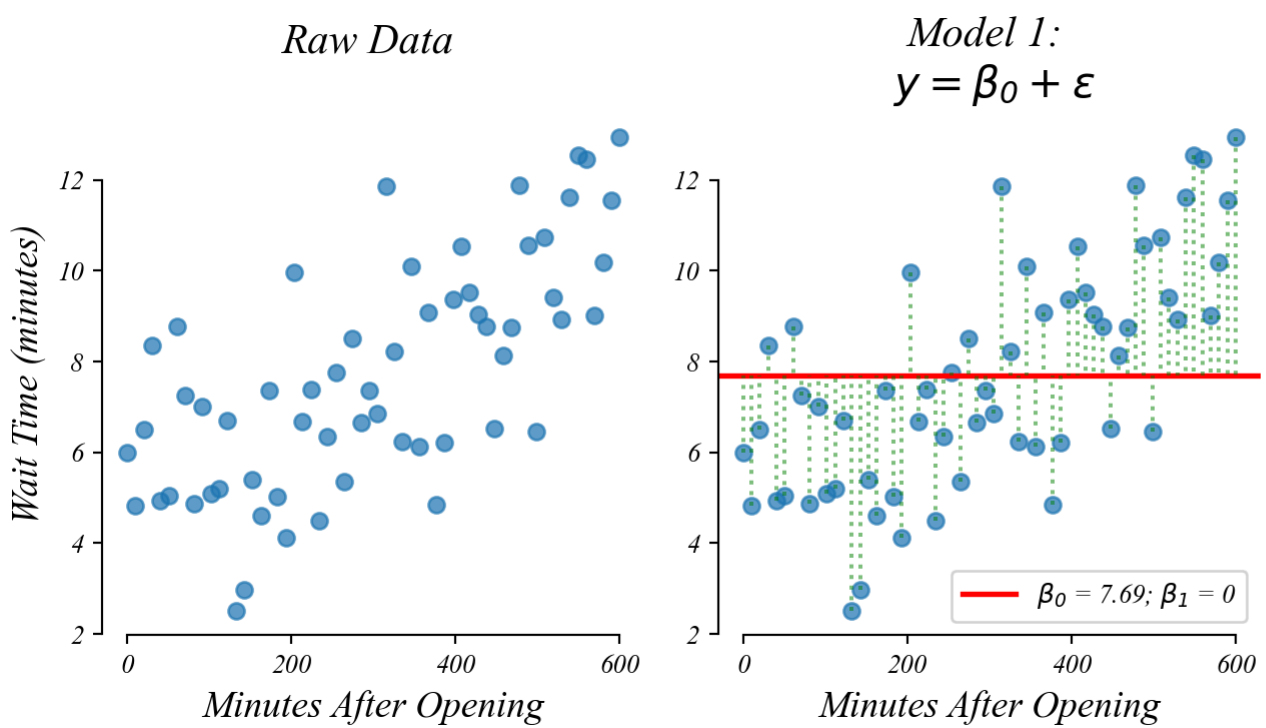
# GLM: bivariate data

*Do people wait longer later in the day?*



# GLM: bivariate data

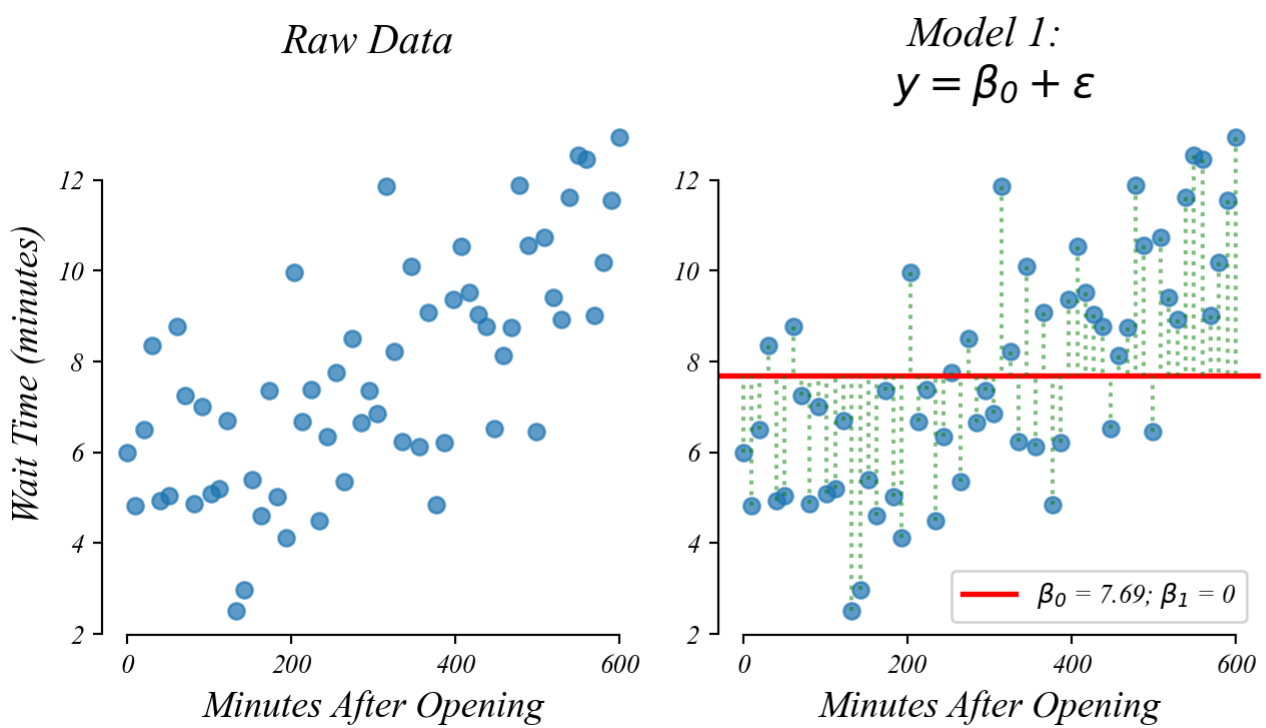
*Do people wait longer later in the day?*




> *but in general we don't ask many questions about vertical incercepts*

# GLM: bivariate data

*Do people wait longer later in the day?*

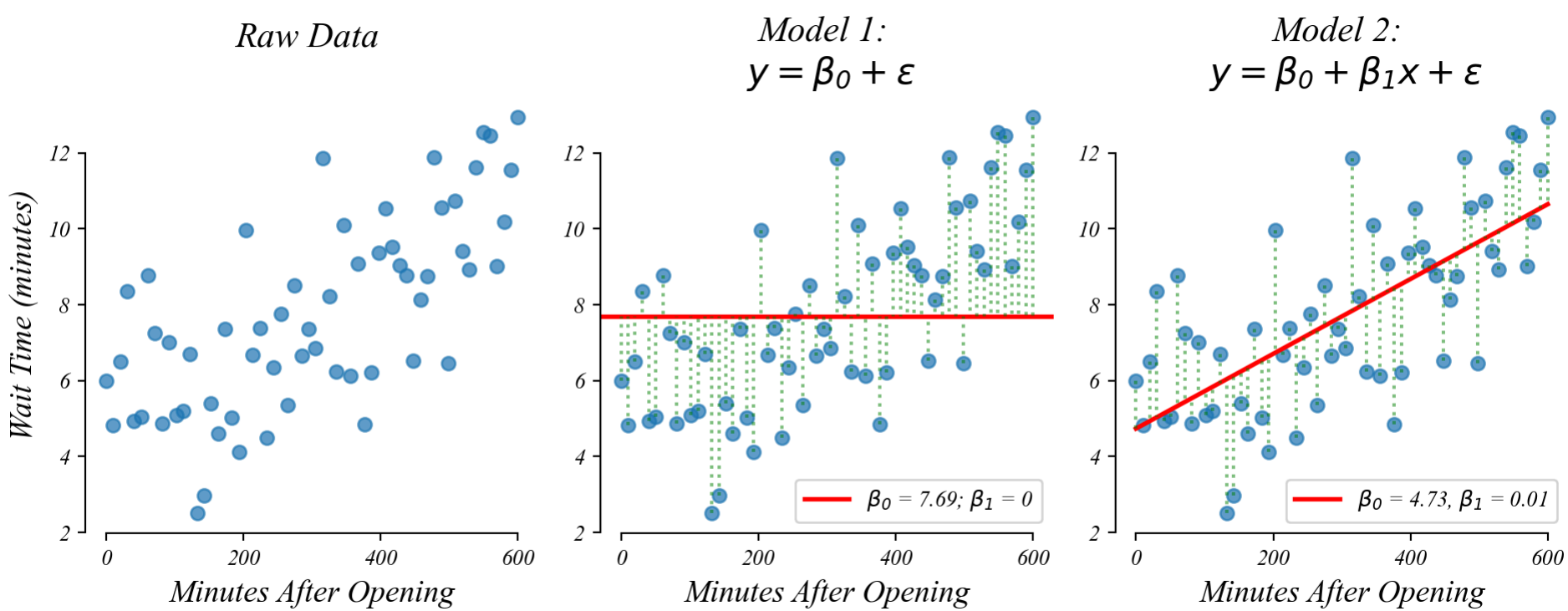



Lets compare two models.

- ***Model 1*** *(Intercept Only):* $y = b$
- ***Model 2*** *(Intercept+Slope):* $y = mx + b$

# GLM: bivariate data

*Do people wait longer later in the day?*

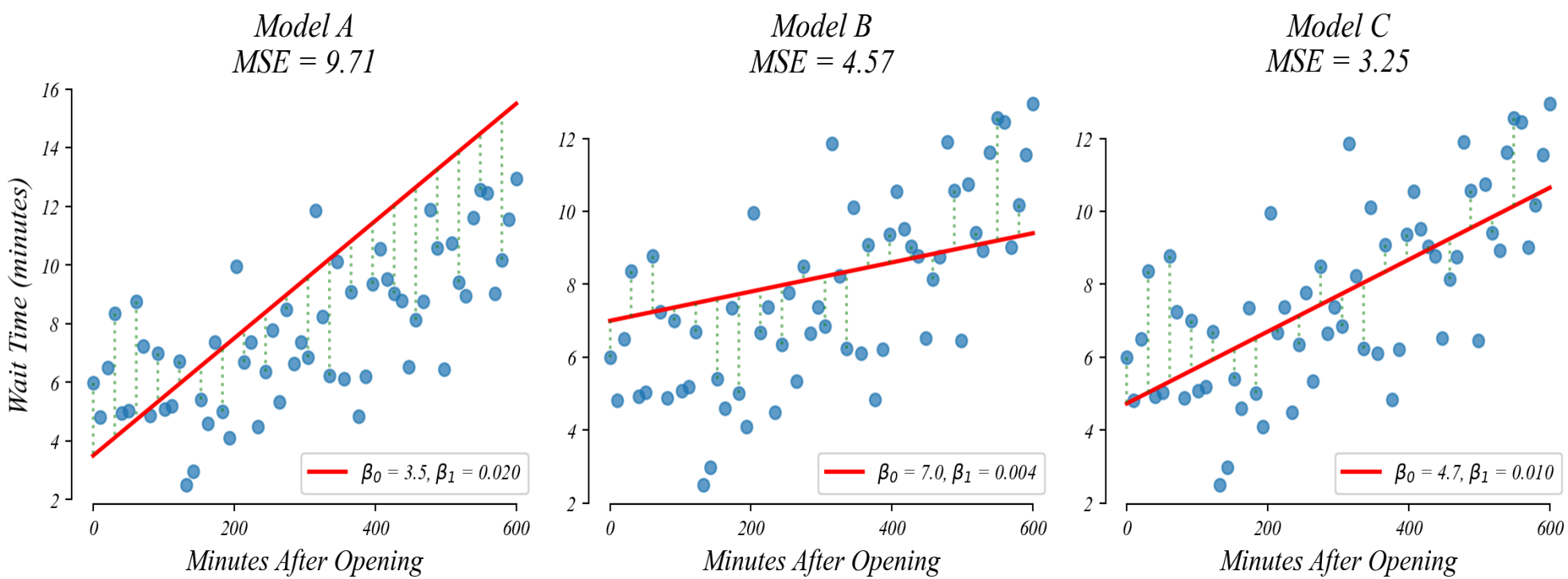



> *a slope ($\beta_1$) improves model fit (MSE; 'wrongness') when there's a relationship*

> *the intercept is no longer the mean*

# Bivariate GLM: minimizing MSE

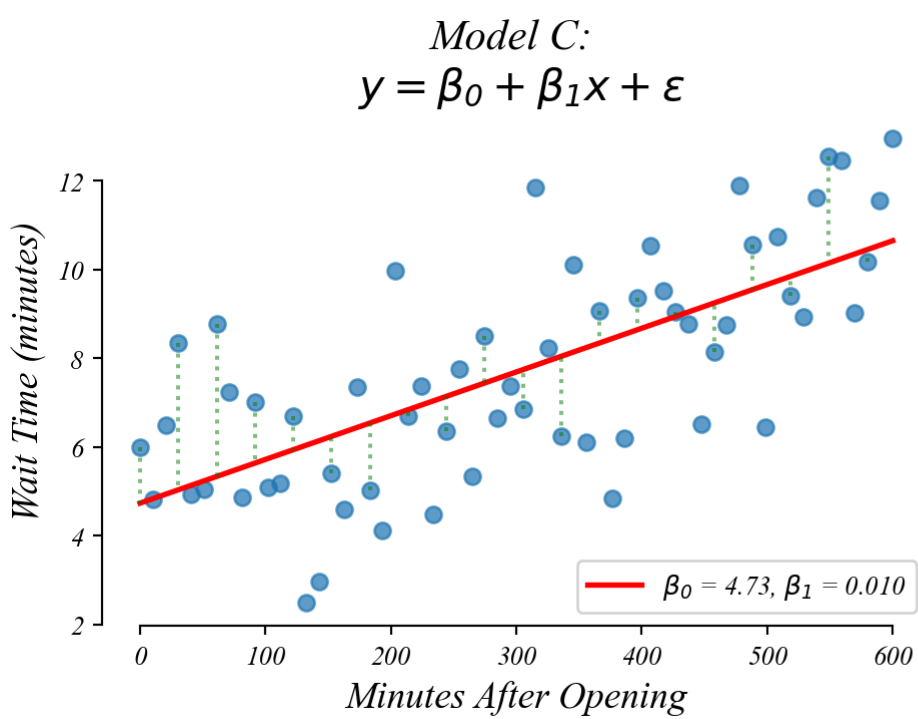
*Which model minimizes the models' 'wrongness' (Mean Squared Error)?*

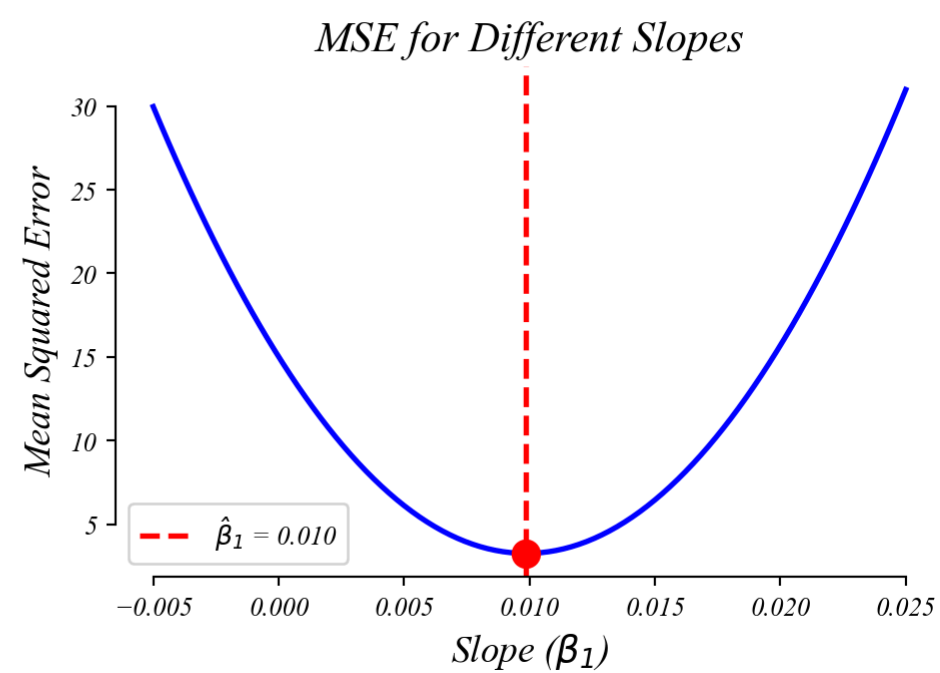



> *Model C minimizes MSE!*

# Bivariate GLM: minimizing MSE

*GLM selects the $\beta_1$ with the smallest MSE.*



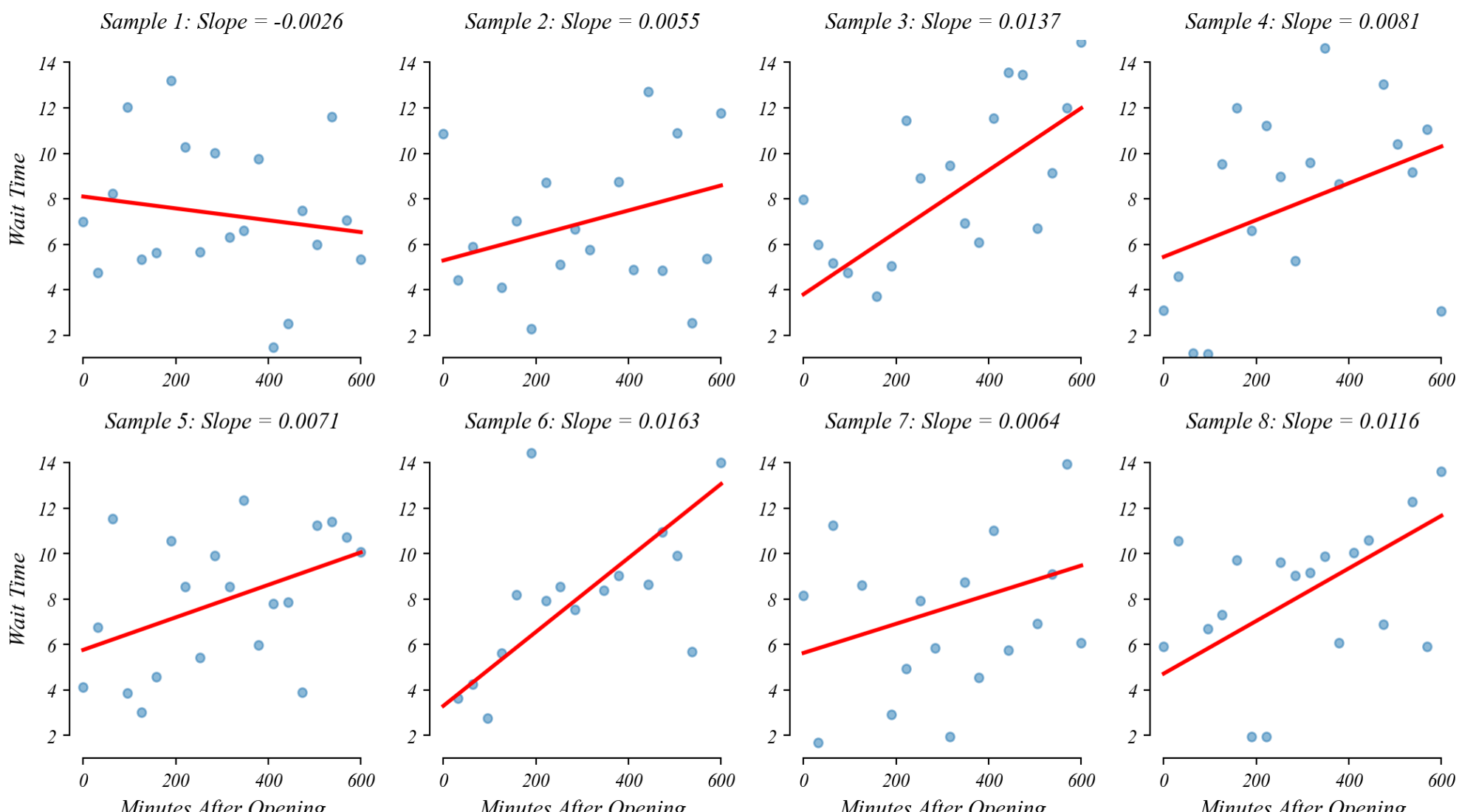




> *this slope ($\beta_1$) gives the best guess of the relationship between x and y*

> *but what if the true slope is zero … could this slope be just sampling error?*

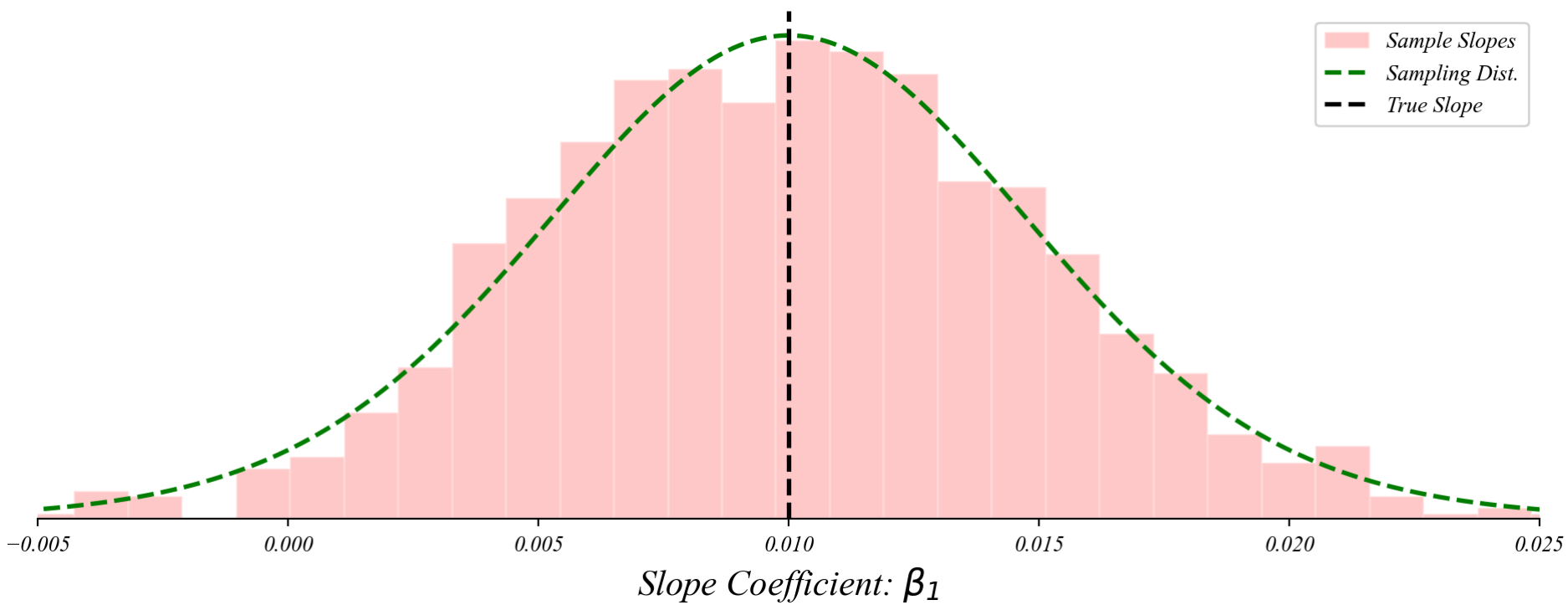
# Bivariate GLM: sampling error

*Like before, if we take many samples, we get slighly different slopes and slighly different fits.*



# Bivariate GLM: sampling distribution of slopes

*The slope coefficient follows a normal distribution centered on the population slope.*

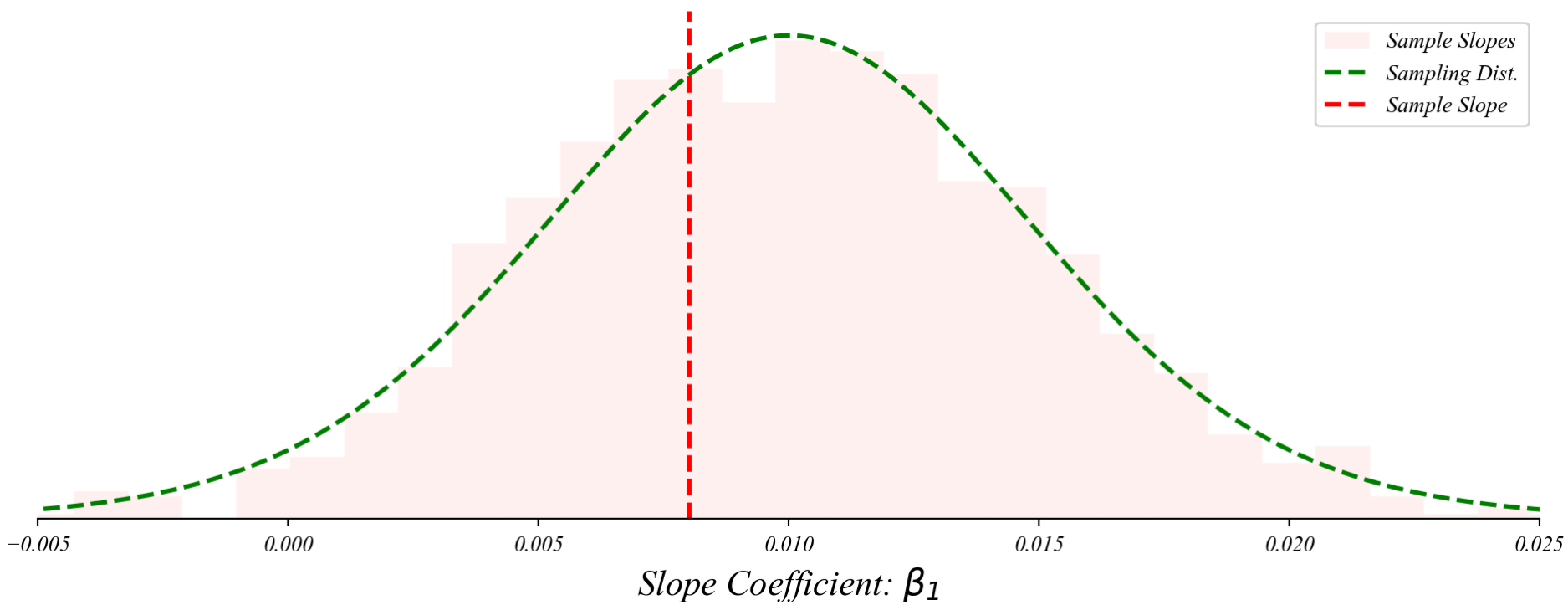



> *the slopes follow a normal distribution around the population relationship!*

> *this lets us perform a t-test on the slope!*

# Bivariate GLM: sampling distribution of slopes

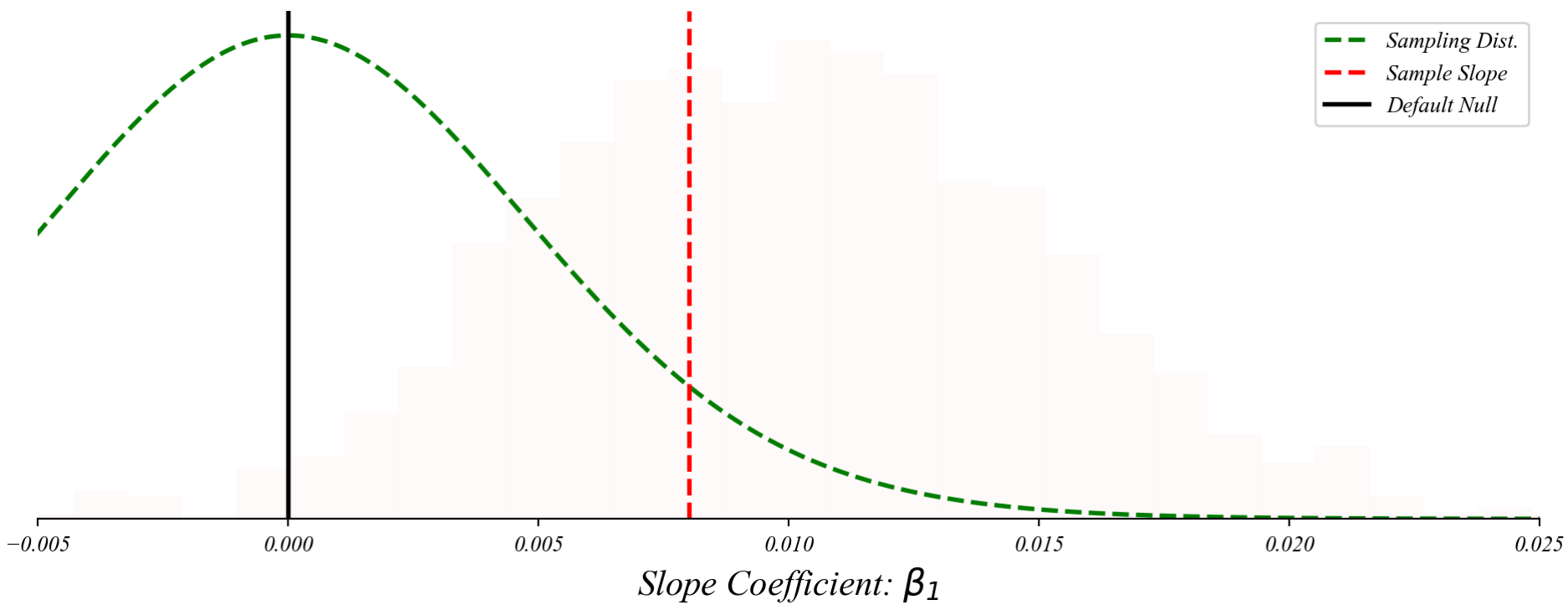
*The slope coefficient follows a normal distribution centered on the population slope.*




> *we don't know the entire distribution, just our sample slope*

# Bivariate GLM: sampling distribution of slopes

*The slope coefficient follows a normal distribution centered on the population slope.*

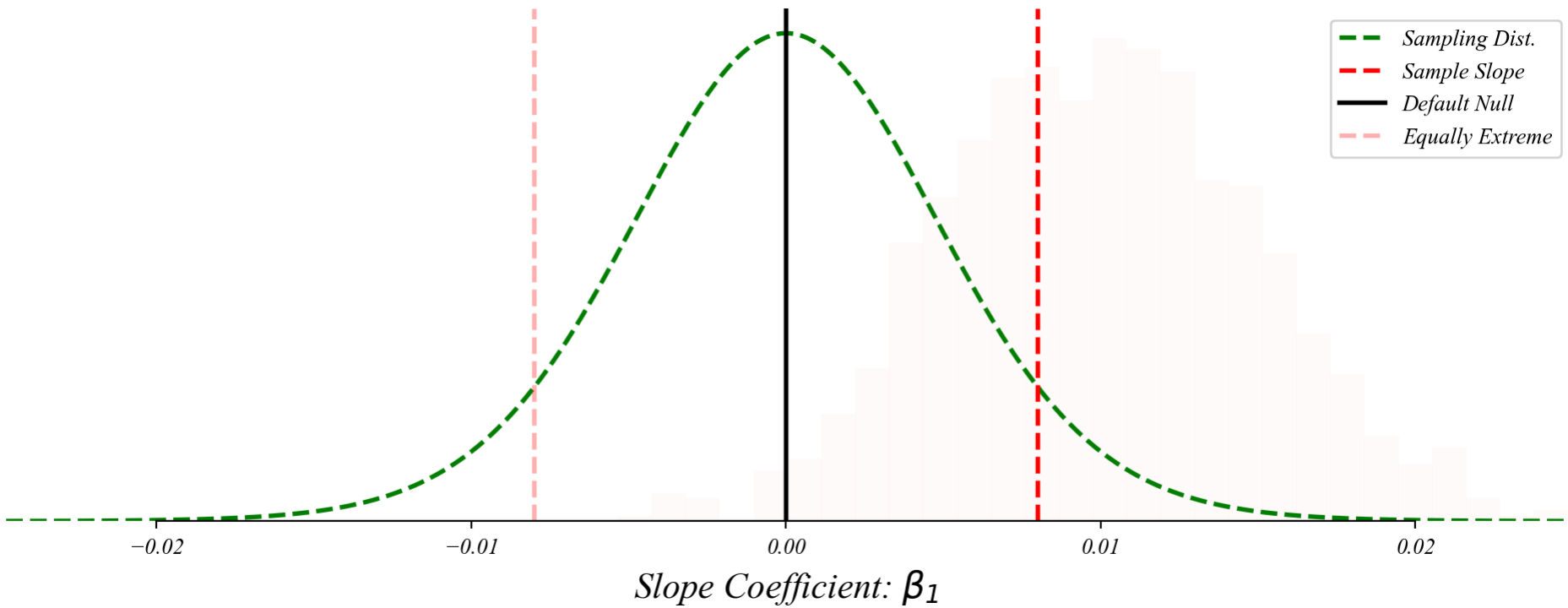



> *center the distribution on our null*

> *check the distance from the sample*

# Bivariate GLM: sampling distribution of slopes

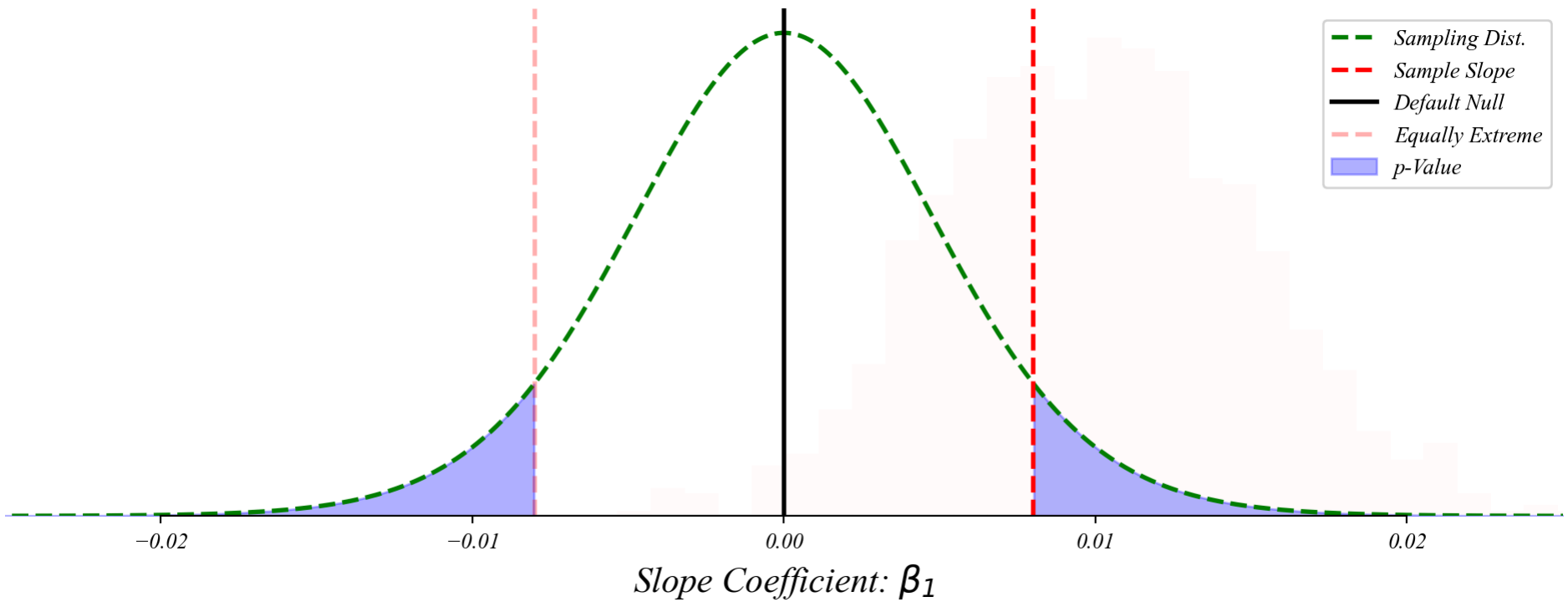
*The slope coefficient follows a normal distribution centered on the population slope.*




> *the p-value is the probability of something as far from the null as our sample*

# Bivariate GLM: sampling distribution of slopes

*The slope coefficient follows a normal distribution centered on the population slope.*



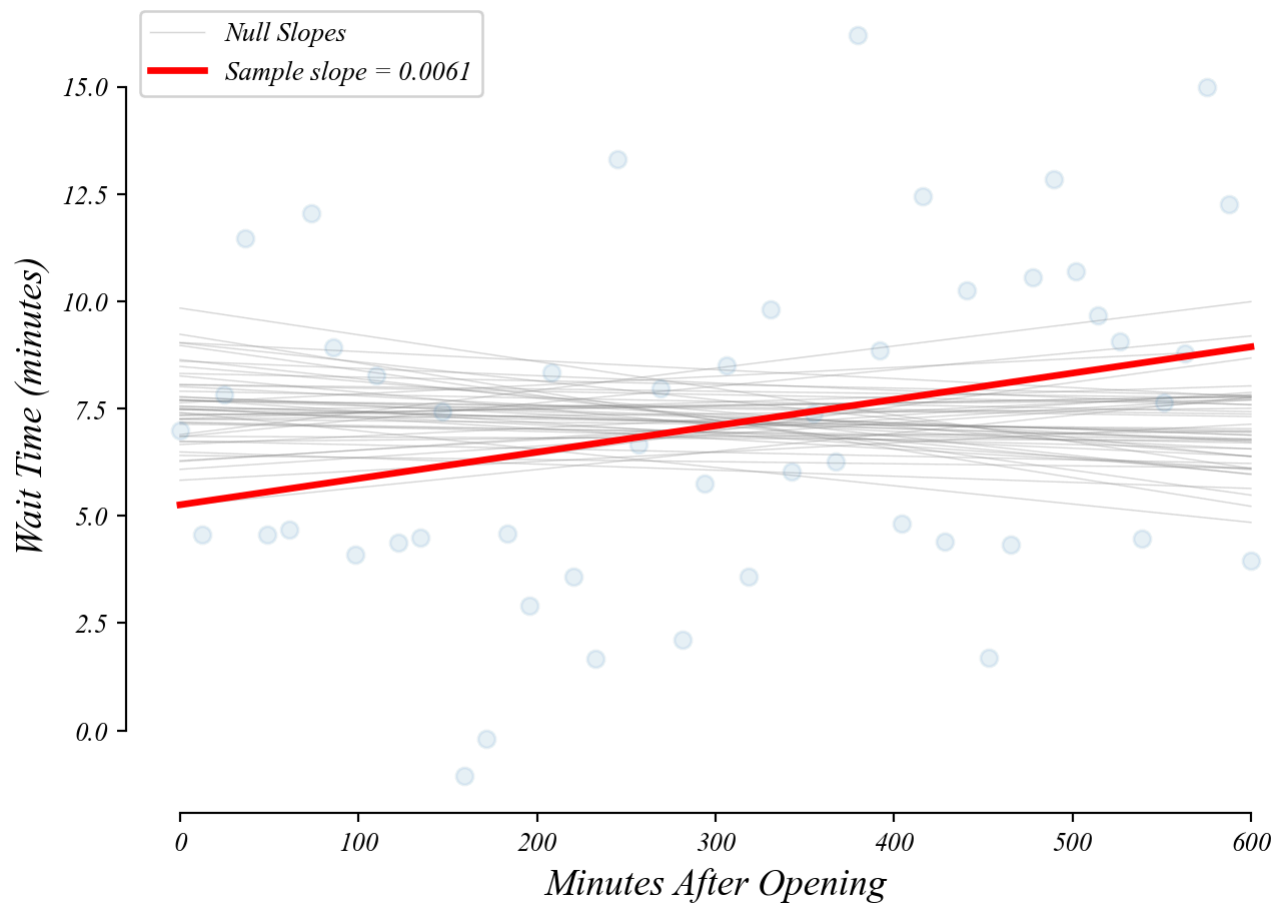

> *p-value: the 'surprisingness' of our sample if* $\beta_1 = 0$

> *the probability of seeing our sample by chance if there is no relationship*

> *a small p-value is evidence against the null hypothesis* ($\beta_1 = 0$)

# Bivariate GLM: sampling distribution of slopes

*Many possible models we might observe by chance if the null ($\beta_1 = 0$) were true.*




> *how likely does it look like this slope was drawn from the null slopes?*

> *p-value: the probability a slope as extreme as ours under the null ($\beta_1 = 0$)*

# Exercise 4.1 | Happiness and Per Capita GDP

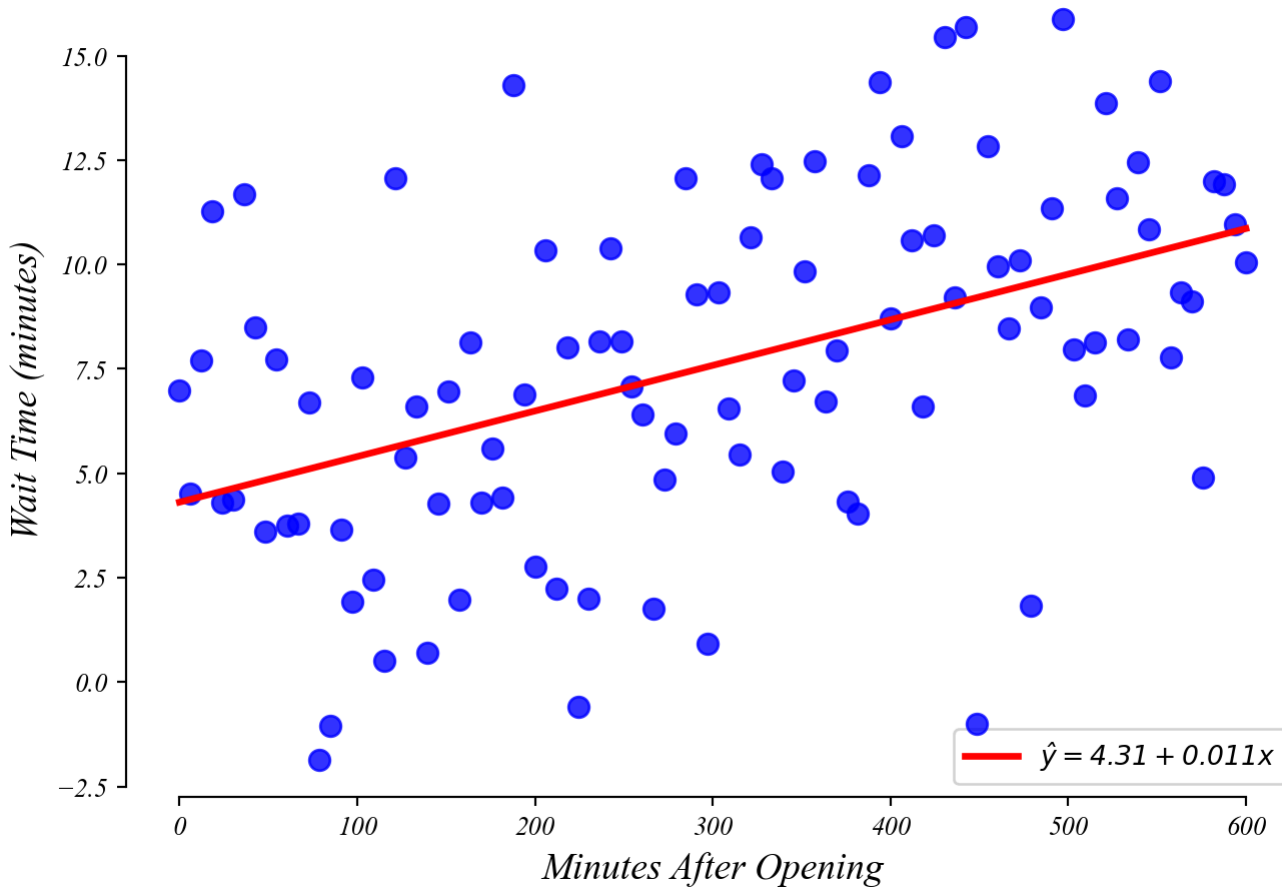
*Are wealtheir countries happier?*

```
# Model: y = b + mx
model = smf.ols('Life_Evaluation ~ log_GDP_PerCap', data).fit() # Intercept is included by def
print(model.summary().tables[1])
```

```
# Visualize: Numerical x Numerical
sns.regplot(data=data, y='Life_Evaluation', x='log_GDP_PerCap')#, ci=None), line_kws={'color':
```

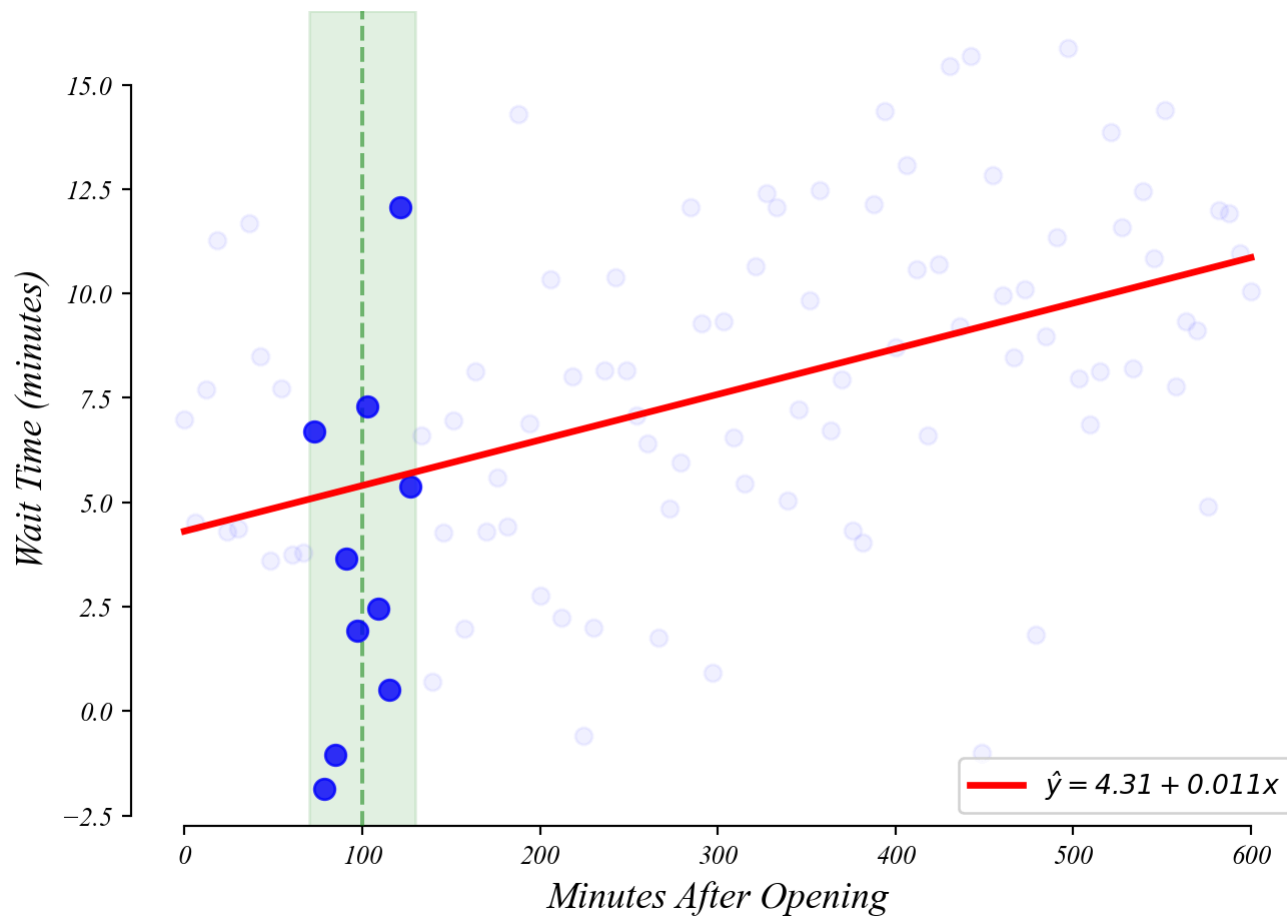
# GLM: predictions

*What wait time should we expect at 100 minutes after open?*



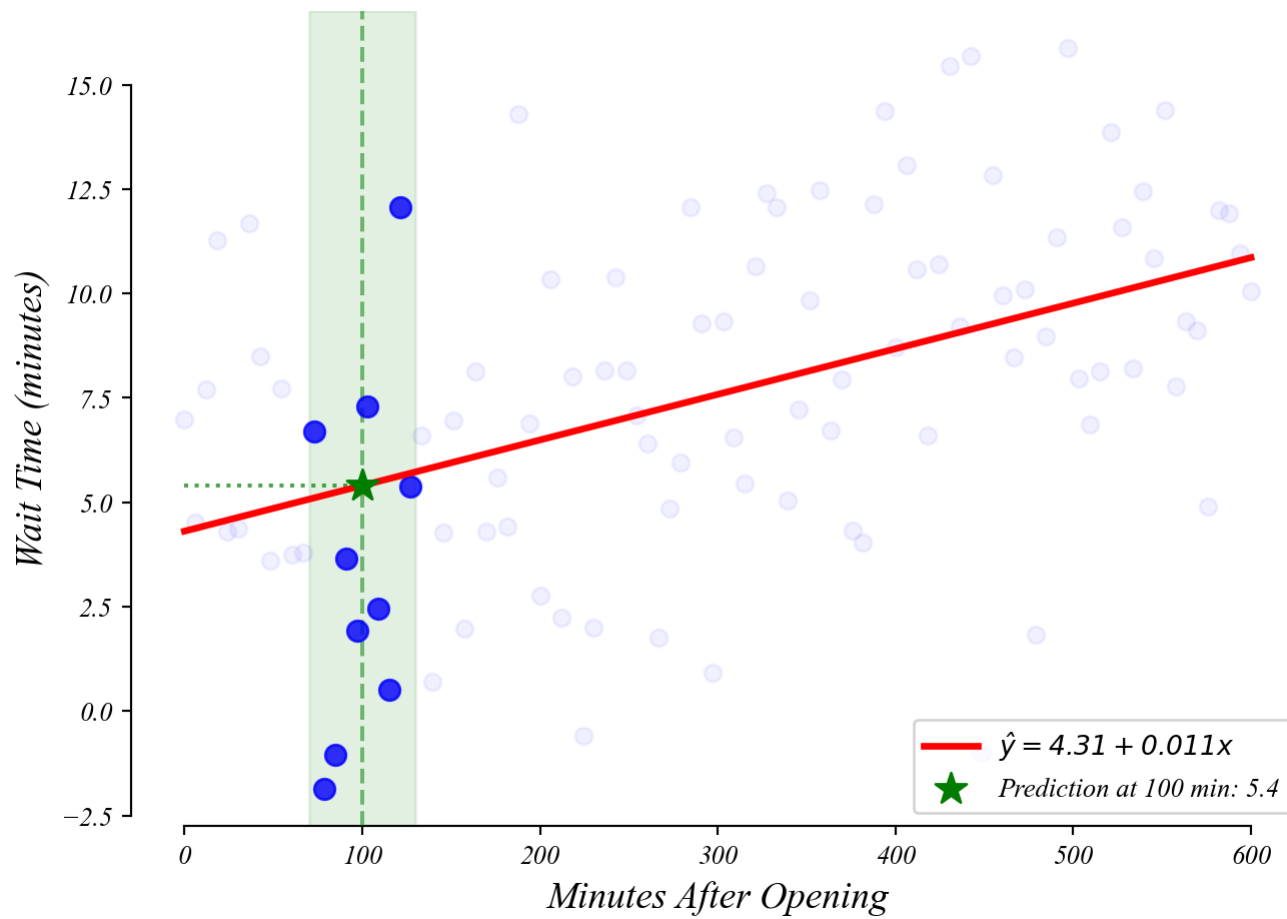
# GLM: predictions

*What wait time should we expect at 100 minutes after open?*



# GLM: predictions

*What wait time should we expect at 100 minutes after open?*

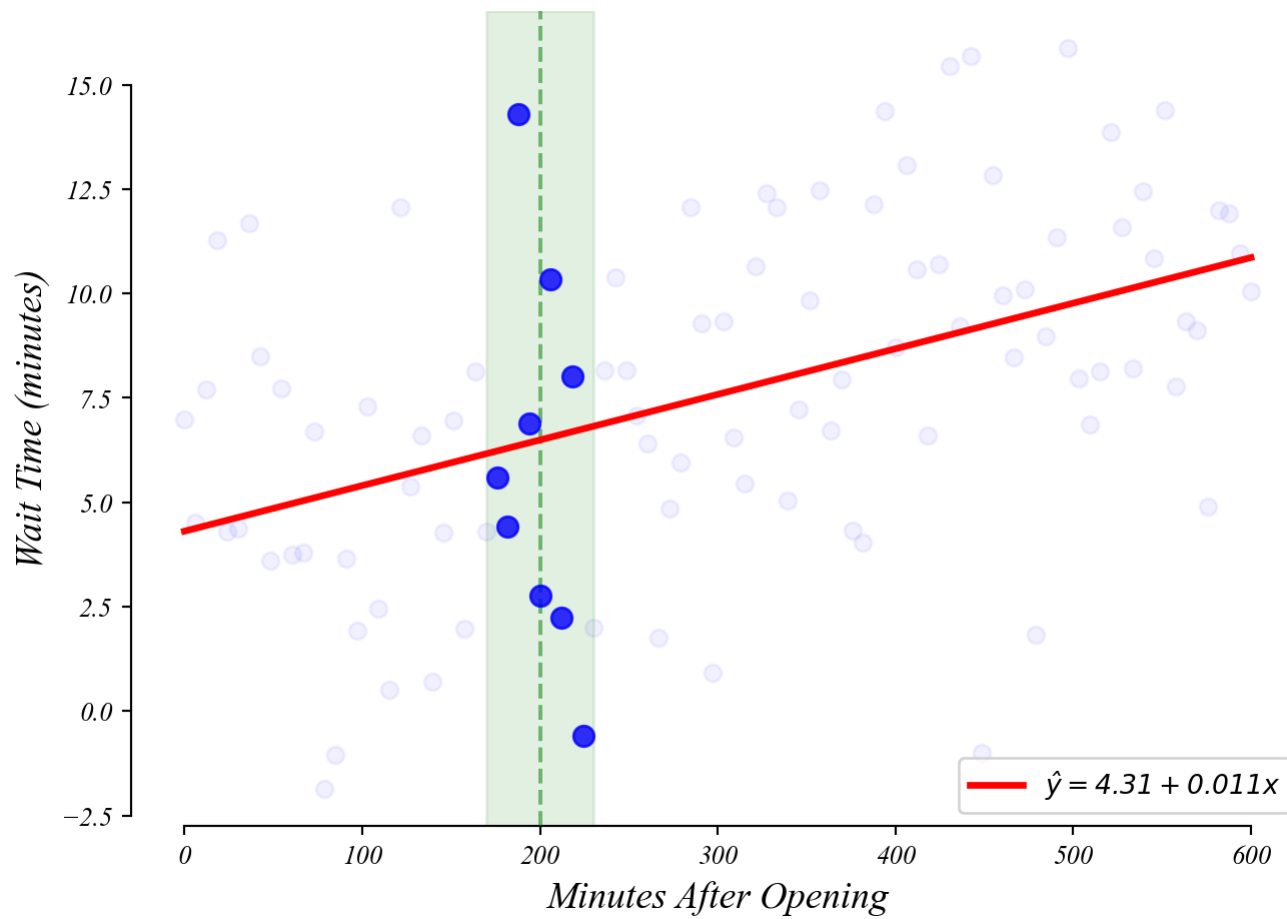



> *you can find this with a calculator!*

> *plug* $x = 100$ *into the equation* $y = 4.31 + 0.011x$

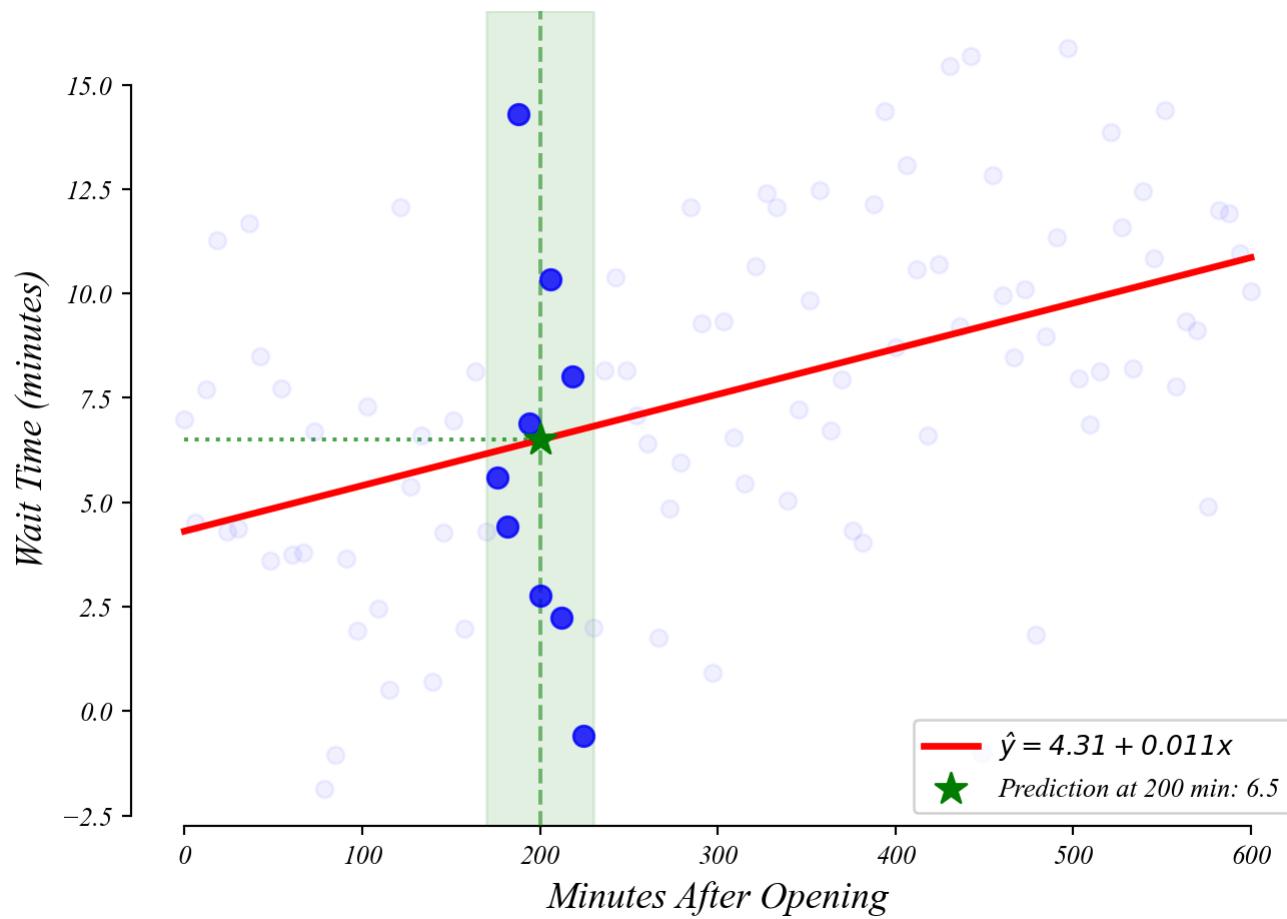
# GLM: predictions

*What wait time should we expect at 200 minutes after open?*



# GLM: predictions

*What wait time should we expect at 200 minutes after open?*



# Exercise 4.1 | Happiness and Per Capita GDP

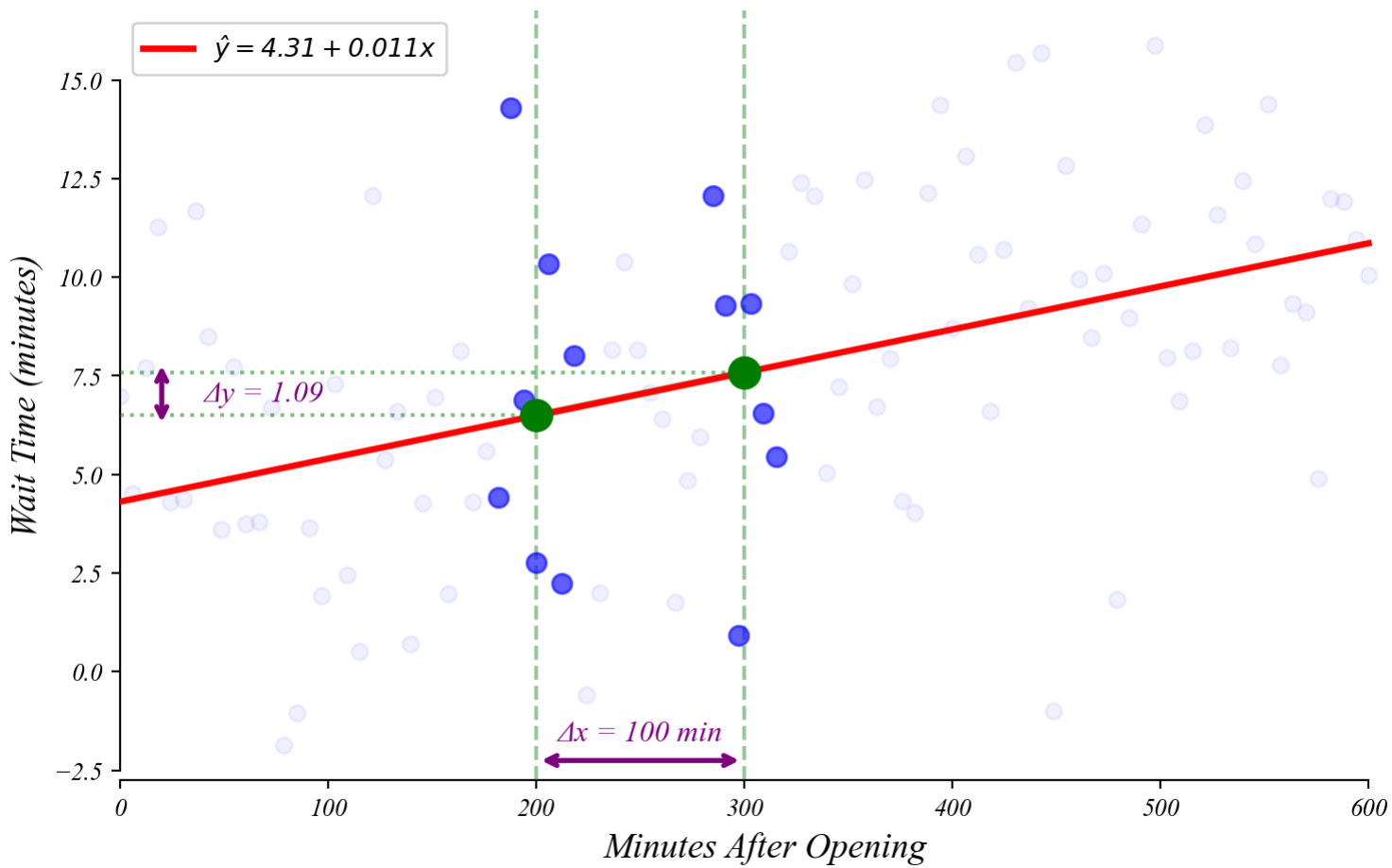
*Are wealtheir countries happier?*

```
# Parameters
b0, b1 = model.params  # intercept, slope
```

```
# Predict: Log GDP of 3
prediction = b0 + b1 * 3
print(prediction)
```

# GLM: interpretation

*How much does wait time increase every minute after open?*




> $\beta_1$ *tells us how much* $y$ *increases with every 1 unit increase in* $x$

# Exercise 4.1 | Happiness and Per Capita GDP

*How much does happiness increase for each additional $1,000 of per capita GDP?*

# The General Linear Model

*Extending the GLM framework*

- *Part 3 | Intercept-Only Model* | $y = \beta_0 + \epsilon$
- *Part 4.1 | Numerical Predictor* | $y = \beta_0 + \beta_1 x + \epsilon$
- *Part 4.2 | Categorical Predictor* | $y = \beta_0 + \beta_1 x + \epsilon$
- *Part 4.3 | Model Diagnostics*
- *Part 4.4 | Causality*

# Looking Forward

*Extending the GLM framework*

Part 5 | Control Variables

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \varepsilon$$

- *Part 5.1 | Numerical Controls*
- *Part 5.2 | Categorical Controls*
- *Part 5.3 | Interactions*
- *Part 5.4 | Model Selection*

> *all built on the same statistical foundation*